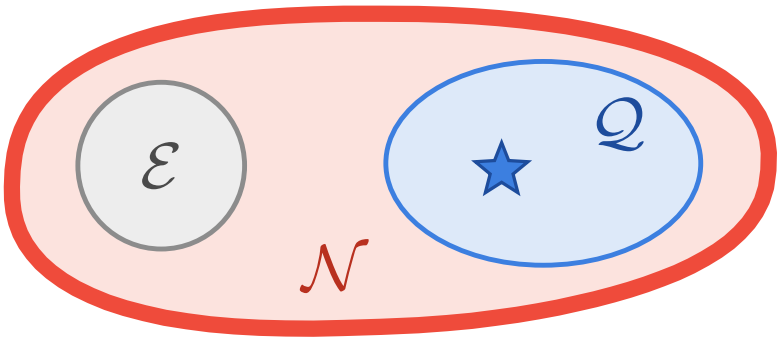
$\mathcal{N}$ noisy quantum systems removing $\leq k$ bits of entropy per step

$\mathcal{Q}$ quantum reservoirs $\mathcal{E}$ measurement-free QEC

a

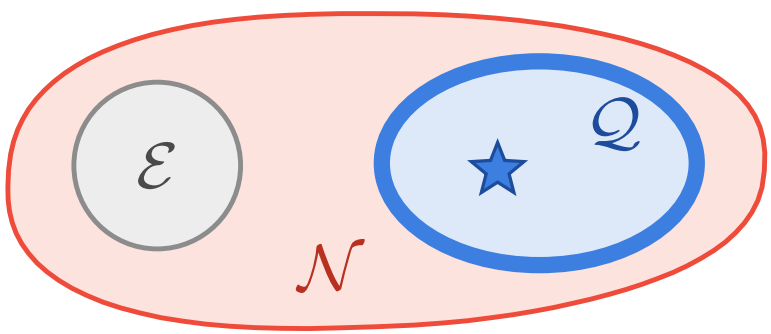


Upper bound

$$\max_{\mathcal{N}} W \leq e^{O(k)}$$

any size, any dynamics

b

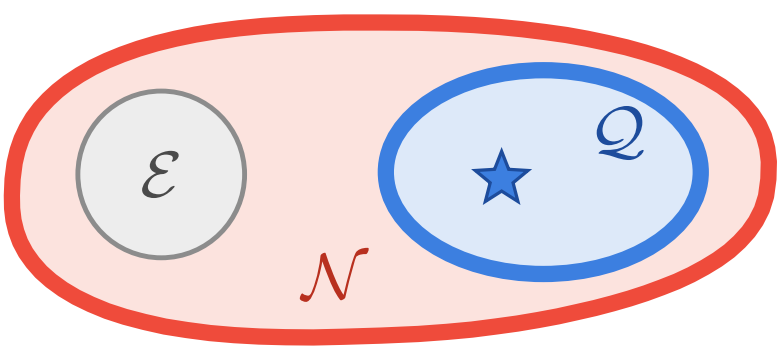


Lower bound

$$\max_{\mathcal{Q}} W \geq e^{\Omega(k)}$$

reached by an explicit reservoir (★)

c



Together: optimal scaling

$$e^{\Omega(k)} \leq \max_{\mathcal{Q}} W \leq \max_{\mathcal{N}} W \leq e^{O(k)}$$

both maxima grow as $e^{\Theta(k)}$